👑 MEP Calculator — Super Admin System Operations Manual

> **IMPORTANT:** This manual is strictly intended for **Super Admins (Application Provider & System Administrators)**. It details global tenant management, license key generation, seat allocation, database backup restoration, and system audit logging.

🗂️ Table of Contents

1. [🏢 Global Organization Management](#-1-global-organization-management) 2. [🔑 License Key Generation & Seat Allocation](#-2-license-key-generation--seat-allocation) 3. [💾 Database Backup, Restoration & Emergency Recovery](#-3-database-backup-restoration--emergency-recovery) 4. [🛡️ Global System Audit Logs & Security Control](#-4-global-system-audit-logs--security-control) 5. [📦 Release & Desktop Installer Management](#-5-release--desktop-installer-management)

🏢 1. Global Organization Management

As a Super Admin, you manage all customer organizations using the **Super Admin Dashboard** (*http://localhost:8081/admin/super*):

``` mermaid flowchart TD Step1["1. Customer Purchases Subscription"] Step2["2. Super Admin Clicks 'Create Organization'"] Step3["3. Enter Org Name & Seat Limit (max_users)"] Step4["4. Generate Organization License Key"] Step5["5. Provide License Key to Org Admin"]

Step1 --> Step2 --> Step3 --> Step4 --> Step5 ```

### Creating & Editing Organizations:

1. Open **Super Admin Dashboard ➔ Organizations Tab**. 2. Click **"Create Organization"**. 3. Input: - **Organization Name**: e.g., *ABC Engineering Consultants LLC* - **Subscription Plan**: e.g., *Pro Enterprise* - **Max Users Seat Limit (** **max_users** **)** : e.g., 10 seats 4. Click **Save**. The system assigns a unique
```

Organization UUID.

🔑 2. License Key Generation & Seat Allocation

Generating Standalone & Organization License Keys:

1. Go to **Super Admin Dashboard ➔ License Management**. 2. Click **"Generate License Key"**. 3. Select License Parameters: - **Bound Organization**: Select target organization. - **Duration / Expiration**: e.g. 365 Days. - **Max User Capacity**: e.g. 10 users. 4. Click **Generate Key**. 5. Copy the generated key (format: MEP-XXXX-XXXX-XXXX) and send it to the customer Organization Admin.

Revoking & Managing License Keys:

- To revoke a key immediately, click **"Revoke"** next to the license key. The system will log out all bound devices on their next 30-second polling cycle.

💾 3. Database Backup, Restoration & Emergency Recovery

Manual Backup Generation:

- Click **"Backup Database Now"** in the Admin Control Panel.
- The server exports a complete timestamped JSON dump (mep_backup_YYYY-MM-DD.json) containing all users, projects, teams, licenses, and logs.

Atomic Database Restore:

1. Click **"Restore Database"**. 2. Select your timestamped backup JSON file. 3. Click **Hydrate & Restore**. The backend will overwrite database tables and clear in-memory caches atomically.

🛡️ 4. Global System Audit Logs & Security Control

- Navigate to **Super Admin Dashboard ➔ Global Audit Logs**.
- Filter activity logs by:

  - **Action Type**: REGISTER, LOGIN, CREATE_ORGANIZATION, ADMIN_DELETE_USER, SAVE_VERSION. - **Date Range / Organization Name.**
- Super Admins can inspect activity logs across all organizations globally.

📦 5. Release & Desktop Installer Management

To publish a new desktop build update: 1. Update version in package.json , tauri.conf.json , Cargo.toml , and update.json to e.g. "0.0.4" . 2. Run npm run tauri:full to compile the installer executable (MEP Calculator_0.0.4_x64-setup.exe). 3. Deploy the updated update.json` manifest to your cloud server. Installed desktop apps will display the **Auto-Update Notification Modal** on launch!